Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/14/2024
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Exploring Spaceflight-Linked Changes in Plant Defense Capabilities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2025
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2025 per the PI (Ed., 6/4/25) NOTE: End date changed to 10/25/2024 per NSSC information (Ed., 10/16/23) NOTE: End date changed to 10/25/2023 per NSSC information (Ed., 11/3/22) NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 10/25/2021 per G. Massa/KSC; forthcoming in NSSC (Ed., 9/24/20)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K0126		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Analysis of the plant biology transcriptomics and proteomics data in the GeneLab data system reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins in space. Anecdotally, plants have also been seen to suffer from pathogen attack during spaceflight missions. Taken together, these observations suggest a hypothesis that plant defense responses may be compromised by spaceflight. Such a disruption of defense would have important implications for the management of plants grown as part of an astronaut life support system. This is especially relevant given the large microbial community on board the International Space Station (ISS) and the possibility that spaceflight triggers enhanced microbial pathogenicity. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. Use of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware growing Arabidopsis seedlings allows direct testing of this idea. Specifically, the project will: (1) Capitalize on the unique capabilities of the BRIC-LED hardware to analyze the transcriptome and cell wall structure responses of Arabidopsis seedlings grown in spaceflight. Plants will be grown in the hardware and then challenged with the bacterial elicitor peptide flg22. Comparison of subsequent on-orbit responses in wild-type and mutants in the defense regulators AtRBOHD and CML24 will then be made. (2) Use comparisons to ground-based control experiments and clinostat and hypoxia treatment to ask whether such changes are likely linked to the microgravity or hypoxic elements of spaceflight and whether these changes may affect the ability of the plant to defend itself against pathogen challenge. The BRIC-LED/Petri Dish Fixation Unit (PDFU) hardware is well-suited to this experimental approach. Thus, the two-stage PDFU hardware will allow both stimulation (injection of flg22 elicitor onto the seedlings) and subsequent fixation of samples (with RNAlater). As the experiment would use flg22 peptide to elicit defense response, potential problems with flying live pathogen will be circumvented. The experimental design will capitalize on the minimal crew time needed for a BRIC-based experiment and the unique capabilities of the BRIC hardware to treat plants in situ on orbit. The results from this experiment will allow direct testing of the critical question of whether the ability of plants to mount a defense response is compromised during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how pathogen sensing and response operates in plants, targeting the Ca2+-dependent and reactive oxygen signaling machinery. The results from this work are not only providing insight into how pathogen signals impact plant growth and defense responses in space but also are contributing to our understanding of similar responses on Earth. Pathogen attack is a major limit on crop productivity with estimates of global losses in the range of 20% across multiple major crops such as wheat, maize, rice, soybean, and potato. The pathogen sensing and response systems triggered by the molecular signals of bacterial attack in space are also those employed by plants challenged by pathogens on Earth. Thus, this work aimed at analysis of how reactive oxygen species and Ca2+ signaling is involved in the defense response systems induced during spaceflight should also help our understanding for these same systems in this general plant defense response system on Earth. These results will provide information to highlight novel molecular targets for potential manipulation to tailor plant responses to the wide spectrum of pathogens that attack plants.
	OVERVIEW Analysis of the available plant biology transcriptomics and proteomics data reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins. These observations suggest plants may exhibit altered defense responses in the spaceflight environment, which would have important practical implications for using plants as part of a bioregenerative life support system. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. This Biological Research in Canisters-Light Emitting Diodes (BRIC-LED)-based research seeks to answer this question using Arabidopsis thaliana as the model plant species. Wild type Arabidopsis and mutants in two defense-related genes will be grown in the BRIC-LED hardware and treated with the bacterial elicitor flg22 to trigger defense signaling. Samples will then be analyzed for alterations in patterns of gene expression linked to defense signaling by comparison to ground controls. The results from this experiment should directly address the critical question of whether plant defense responses are compromised during spaceflight. PROGRESS 1. The flight component of this project has been awaiting the availability of the BRIC-LED hardware and an associated flight opportunity. After a successful functional test, the BRIC-LED facility is now ready for use and an Experiment Verification Test (EVT) has been scheduled (currently for October 2024). Fresh seed stocks have been maintained for all the genotypes for the experiment and are regularly being tested for germination and growth characteristics to be ready for the EVT and a potential flight in 2025. 2. In parallel to preparing for the flight, experiments have been ongoing to characterize how spaceflight conditions such as hypoxic stress might be influencing plant defense capabilities, using a combination of ground-based studies and analysis of previous spaceflight data. Spaceflight is thought to impose hypoxic conditions on biology due to the loss of convection in the microgravity environment. Bakshi et al., (2023) details an analysis of ground-based studies of mutants of Arabidopsis thaliana where the gene for the CAX2 vacuolar Ca2+ transporter had been knocked out. This work showed these mutants were resistant to waterlogging stress, likely due to constitutive activation of their natural hypoxia tolerance system. Previously, CAX2 knockouts have been flown in the Advanced Plant Experiments-05 (APEX-05) flight experiment, where Arabidopsis plants were grown on the International Space Station. The data from this flight experiment have been further investigated and have revealed a potential link between spaceflight-induced hypoxic stress and altered plant defense capabilities. In these APEX-05 plants, wild type showed an inhibition of root and shoot growth in space, whereas CAX2 knockouts already exhibited this phenotype on the ground and showed no further effects when grown in spaceflight. Transcriptome analyses, using both RNA sequencing (RNAseq) and quantitative polymerase chain reaction (PCR), indicate that the wild type plants exhibited molecular signatures of having experienced hypoxia and oxidative stress while growing in space. These same plants showed a disruption in the expression of genes related to defense signaling in both roots and

Task Progress:	shoots. Neither the hypoxic signature, nor the defense disruption were observed in the CAX2 mutant lines, where alterations in cell wall-related transcripts in the shoots was the only significant response. Further investigation of the ground controls for this experiment shows that the CAX2 mutants showed down-regulation of both hypoxic response and defense-related genes on the ground, in comparison to wild type plants also grown on the ground. These observations suggest knocking out CAX2 constitutively suppresses both hypoxic and defense responses, implying these pathways may be linked. Indeed, flooding stress and its related hypoxic conditions are now recognized as likely leading to shifts in defense capabilities (García et al., 2024) with, e.g., hypoxia-linked defense priming thought to ready the plant for increased pathogen attacks associated with flooding and post-flooding stresses (Hsu et al., 2013). Conversely, flooding has been proposed to suppress the jasmonic acid-related anti-herbivory network (Rehman et al., 2023), implying a broad spectrum of effects on the plant's ability to defend itself. Ground-based experiments growing wild type and CAX2 mutants in chambers, where the ambient oxygen levels can be controlled, indicate these changes seen in spaceflight are mimicked at ~6% oxygen levels. This level of hypoxia is consistent with previous research, where a similar level of hypoxic stress can copy the effects of spaceflight on the oxidative stress responses in Arabidopsis (Choi et al., 2019). Current work is using ground-based experiments to ask if this level of hypoxic stress leads to a compromised ability to trigger immune responses and whether the CAX2 mutants are less susceptible to this effect. REFERENCES CITED Bakshi, A., Choi, W.-G., Kim, S.-H., and Gilroy, S. (2023). The vacuolar Ca(2+) transporter CATION EXCHANGER 2 regulates cytosolic calcium homeostasis, hypoxic signaling, and response to flooding in Arabidopsis thaliana. New Phytol. 240: 1830–1847. Choi, W.G., Barker, R.J., Kim, S.H., Swanson, S.J., and Gilroy, S. (2019). Variation in the transcriptome of different ecotypes of Arabidopsis thaliana reveals signatures of oxidative stress in plant responses to spaceflight. Am. J. Bot. 106: 123–136. García, P., Singh, S., and Graciet, E. (2024). New Insights into the Connections between Flooding/Hypoxia Response and Plant Defenses against Pathogens. Plants 13. Ge, S.X., Son, E.W., and Yao, R. (2018). iDEP: An integrated web application for differential expression and pathway analysis of RNA-Seq data. BMC Bioinformatics 19: 534. Hsu, F.C., Chou, M.Y., Chou, S.J., Li, Y.R., Peng, H.P., and Shih, M.C. (2013). Submergence confers immunity mediated by the WRKY22 transcription factor in Arabidopsis. Plant Cell 25: 2699–2713. Rehman, M., Saeed, M.S., Fan, X., Salam, A., Munir, R., Yasin, M.U., Khan, A.R., Muhammad, S., Ali, B., Ali, I., Khan, J., and Gan, Y. (2023). The Multifaceted Role of Jasmonic Acid in Plant Stress Mitigation: An Overview. Plants (Basel, Switzerland) 12.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Bakshi A, Choi WG, Kim SH, Gilroy S. "The vacuolar Ca2+ transporter CATION EXCHANGER 2 regulates cytosolic calcium homeostasis, hypoxic signaling and response to flooding in Arabidopsis thaliana." New Phytol. 2023 Dec;240(5):1830-47. https://doi: 10.1111/nph.19274 ; PubMed PMID: 37743731 , Dec-2023
Books/Book Chapters	Chin S, Johns S, Gilroy S. "Calcium in root growth and development." in "Plant Roots: The Hidden Half, Fifth Edition (5th ed.)." Ed. T. Beeckman, A. Eshel. CRC Press, Boca Raton, FL: CRC Press, 2024. Chpt 19. https://doi.org/10.1201/b23126 , Aug-2024